

Question 1



[Maximum mark: 6]

The following transition diagram reflects the proportions of customers that Qatar Airways loses to its competitor airlines each year, and vice versa.



(a) Construct a transition matrix T with elements in decimal form.

[2]

(a)

		From	
		Qatar airways	Competitor
To	Qatar airways	0.75	0.15
	Competitor	0.25	0.85

Ti-nspire CX ii

scratchpad

→ Template



→ Choose matrix of required order

else use → menu → 7 Matrix → 1 Create (enter) → 1 Matrix

No. of rows

No. of column

$$T = \begin{bmatrix} 0.75 & 0.15 \\ 0.25 & 0.85 \end{bmatrix}$$

(b) Interpret the meaning of the elements with values

(i) 0.15

(ii) 0.75

[2]

(i) 15% of customer from Competitor will use Qatar airways next year.

(ii) 75% of Qatar airways customers will retain in Qatar airways next year.

Assume that the initial state of the market share is

$$s_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \rightarrow (100\% \text{ of customers are with Qatar Airways})$$



(c) Determine the market share of Qatar Airways after 5 years.

[2]

(To save)

$$\begin{bmatrix} 0.75 & 0.15 \\ 0.25 & 0.85 \end{bmatrix} \rightarrow \text{ctrl} \rightarrow \text{var}(s_0 \rightarrow) \rightarrow t$$

$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \rightarrow s$ stored in 's'

$$T^5 \cdot s_0 = \begin{bmatrix} 0.75 & 0.15 \\ 0.25 & 0.85 \end{bmatrix}^5 \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$= T^5 \times s = \begin{bmatrix} 0.4236 \\ 0.5764 \end{bmatrix} \quad (\text{using GDC})$$

Hence the market share of Qatar Airways after 5 years will be 42.4%

Question 2

[Maximum mark: 6]

Three Internet Service Providers (ISPs) are available in a small town. During the year, ISP A is expected to retain 85 % of its customers; 10 % will be lost to ISP B and 5 % to ISP C. ISP B is expected to retain 80 % of its customers; 10 % will be lost to each of the other two ISPs. ISP C is expected to retain 75 % of its customers; 15 % will be lost to ISP A and 10 % to ISP B.

(a) Write down a transition matrix that describes the exchange of market shares between the three ISPs during the year.

[2]

		From		
		ISP A	ISP B	ISP C
To	ISP A	0.85	0.10	0.15
	ISP B	0.10	0.80	0.10
	ISP C	0.05	0.10	0.75

scratchpad → Template  → Choose matrix of required order

$$T = \begin{bmatrix} 0.85 & 0.10 & 0.15 \\ 0.10 & 0.80 & 0.10 \\ 0.05 & 0.10 & 0.75 \end{bmatrix} \quad \begin{array}{l} \text{To save} \\ \boxed{} \rightarrow \text{Ctrl} \rightarrow \text{Var(sto)} \rightarrow \\ \text{"X"} \end{array}$$

The current market share held by ISP A is 0.2, by ISP B is 0.3 and by ISP C is 0.5.

(b) Find the market share held by each ISP after one year.

[2]

(c) Find the market share held by each ISP after five years if the same trend of market share exchanges between the three ISPs continues.

[2]

$$(b) \quad S_0 = \begin{bmatrix} 0.2 \\ 0.3 \\ 0.5 \end{bmatrix} \rightarrow \text{"y"}$$

$$S_1 = T^1 \times S_0 = X \cdot Y = \begin{bmatrix} 0.275 \\ 0.310 \\ 0.415 \end{bmatrix}$$

After one year the market share of each ISP is as follows:

ISP A : 0.275

ISP B : 0.310

ISP C : 0.415



After 5 years

$$S_5 = T^5 S_0 = X^5 \times y = \begin{bmatrix} 0.85 & 0.10 & 0.15 \\ 0.10 & 0.80 & 0.10 \\ 0.05 & 0.10 & 0.75 \end{bmatrix}^5 \begin{bmatrix} 0.2 \\ 0.3 \\ 0.5 \end{bmatrix}$$
$$= \begin{bmatrix} 0.405362 \\ 0.327731 \\ 0.266908 \end{bmatrix}$$

Hence after 5 years the market share of each ISP is as follows:

ISP A: 0.405

ISP B: 0.328

ISP C: 0.267

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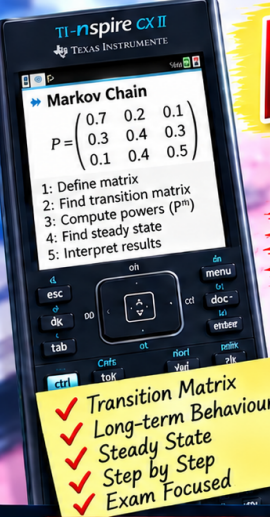
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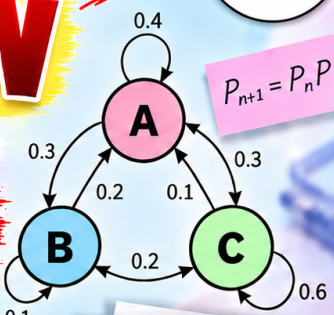
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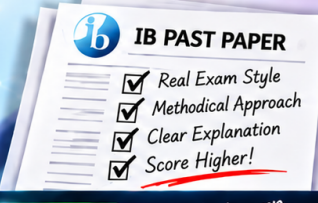
Part-2



Transition Matrix
Long-term Behaviour
Steady State
Step by Step
Exam Focused



$P_{n+1} = P_n P$



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